Shared Structure & Low Rank Representations: Testing Platonic Scaling in Cross-Modal Astronomy

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Abstract

The Platonic Representation Hypothesis predicts increasing representational convergence with model scale [Huh et al., 2024]. We test that prediction by learning a shared sparse basis between independently trained SAE representations of matched Legacy Survey–HSC models [UniverseTBD et al., 2025], so that cross-modal transfer can be compared in a common feature coordinate system rather than only in dense embeddings. The shared-basis codes recover substantially more held-out neighbourhood correspondence than the dense baseline, including on the smallest rungs, but that additional alignment shows little systematic dependence on model size. The same protocol applied to block-sparse featurizers, and a qualitatively different unpaired alignment [Jha et al., 2025], give the same picture: within-family probe×scale interactions are small and mixed in sign (SAE mean $D{=}{-}5.2{\times}10^{-4}$; BSF mean $D{\approx}0$; $6/11$ positive in each case). A shared basis can therefore reveal substantial cross-modal structure without relying on scale, while making the models larger does not reliably increase that structure. Scale does not necessarily imply a more shared representation; raw parameter count is, at best, an incomplete control variable for representational convergence.

1 Introduction

Representations from independently trained models often become more aligned as capability and scale increase [Huh et al., 2024]. The Platonic Representation Hypothesis (PRH) interprets this as convergence toward a shared statistical model of reality [Huh et al., 2024]. In astronomy, matched survey embeddings give a relatively clean test of this idea: previous work reports increasing mutual k-nearest-neighbour (mKNN) alignment with model capacity [UniverseTBD et al., 2025].

There is an immediate ambiguity, however. Measured alignment depends on how the representation is read. Dense mKNN compares neighbourhoods in the native embedding spaces. Sparse autoencoders instead decompose each model into its own dictionary [Gao et al., 2025], so the resulting codes are not a priori comparable. The construction we contribute is a learned shared basis: independently trained TopK SAEs encode the two surveys, and a linear map, fit on paired training objects only, sends one model's codes into the other's dictionary. Cross-modal transfer is then neighbourhood overlap in that shared sparse coordinate system. We apply the same protocol to block-sparse featurizers (BSFs) [Fel et al., 2026]. SAE feature-space correspondence has been studied across language models [Lan et al., 2024]; here the shared basis is used to measure *modal transfer* between survey-specific sparse representations of foundation models.

If that construction exposes substantially more Legacy↔HSC correspondence than the dense embeddings do, does the extra correspondence appear preferentially in the larger models? If not, then scaling a model does not, by itself, imply a more shared representation.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

We test that question directly. On a matched held-out evaluation, shared-basis SAE and BSF increase neighbourhood overlap across every model-size rung, including the smallest. We define the lift

$$L_R(P) = M_R(P) - M_{\mathrm{dense}}(P),$$

and ask whether $L_R(P)$ itself grows with P. It does not in any clear or systematic way. The corresponding within-family interaction,

$$D_R = \Delta M_R - \Delta M_{\mathrm{dense}},$$

is small and mixed in sign.

We then repeat the question with a qualitatively different probe: an unpaired nonlinear alignment inspired by Jha et al. [2025]. It recovers nontrivial relational structure without seeing paired training objects, but again gives no clean monotonic relationship with model size.

The point is therefore not that any particular probe "fails to scale." It is that scale is a weak proxy for shared representation. Changing the probe changes the *level* of measured correspondence much more reliably than it changes the *size response*: smaller models already share transferable sparse structure, and larger models do not systematically share more of it. That distinction is the main result of the paper.

2 Experimental setup and alignment probes

2.1 Legacy↔HSC model-size ladders

We use the official UniverseTBD Legacy Survey [Dey et al., 2019]↔HSC [Miyazaki et al., 2018] matched embeddings [UniverseTBD et al., 2025]. Each rung pairs the same architecture size across surveys; we use n=16,384 matched objects. The model families are AstroPTv2 (15, 95, 850M) [Smith et al., 2024], ConvNeXtv2 (nano–large) [Woo et al., 2023], DINOv2 (small–giant) [Oquab et al., 2024], ViT (base–huge) [Dosovitskiy et al., 2021], and I-JEPA (huge, giant) [Assran et al., 2023]. The unpaired analysis covers ConvNeXt and DINOv2.

2.2 mKNN and matched evaluation

For paired representations A and B,

$$M_k = \frac{1}{n}\sum_i \frac{|\mathcal{N}_k^A(i) \cap \mathcal{N}_k^B(i)|}{k},$$

using cosine neighbours with self-exclusion [Huh et al., 2024, UniverseTBD et al., 2025]. Neighbour lists are sliced to k before intersection. Our primary scale is k=10. Under a uniform-neighbour null, expected overlap is $k/(n-1)$ [Gröger et al., 2026].

We retain the original full-catalog dense score only to connect back to the scaling analysis of UniverseTBD et al. [2025]. All probe×scale comparisons use the same held-out query IDs, full gallery, k, self-exclusion, and object set (n_{test}=3277). Dense, SAE, and BSF are therefore directly comparable in both level and size response.

2.3 Dense embeddings versus a learned shared basis

The dense probe uses the raw ℓ_2-normalised embeddings with cosine mKNN. That comparison never places the two models in a common feature basis: it only asks whether neighbourhoods in the two native spaces overlap.

The SAE probe does place them in a common basis. We encode both surveys with pretrained TopK sparse autoencoders [Gao et al., 2025], then fit a Ridge map from Legacy codes into the HSC dictionary using train IDs only, with train-only IDF weighting (`shared_side1_basis_idf`). The mapping direction is fixed in advance. Held-out mKNN is then computed between the mapped Legacy codes and the HSC codes, i.e. as cross-modal transfer in the shared dictionary rather than as a dense Procrustes of the original embeddings. Dictionary width F=2048 and seed 0 are shared across the ladder; TopK sparsity varies with the available checkpoint ($k \in \{18, \ldots, 23\}$).

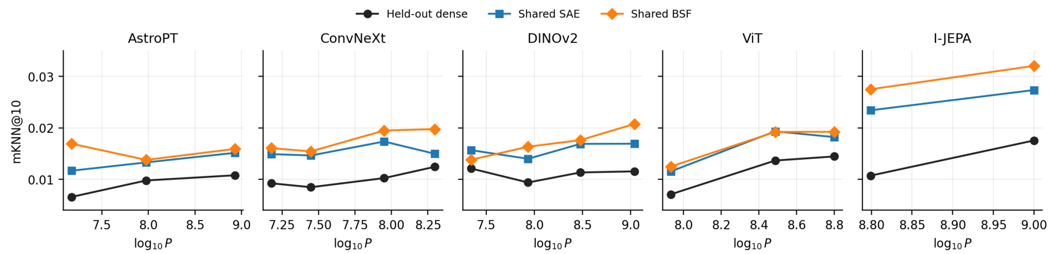

Figure 1: Held-out mKNN@10 across Legacy↔HSC model ladders using identical queries and galleries for dense embeddings, SAE codes mapped into a shared dictionary, and the same protocol for BSF. The shared basis raises the level of correspondence at every rung; the within-family size trajectories change much less.

For BSF, we replace SAE codes with Vanilla BSF full signed codes [Fel et al., 2026] ($G{\approx}512$ blocks) and apply the same shared-basis protocol.

For each within-family adjacent step $P_j \to P_{j+1}$, we compute

$$D_{R,j} = \Delta M_{R,j} - \Delta M_{\text{dense},j} = L_R(P_{j+1}) - L_R(P_j),$$

for $R \in \{\text{SAE}, \text{BSF}\}$. This is the quantity we care about: whether the extra correspondence exposed by the shared basis itself increases with size.

2.4 Unpaired relational probe

Our second analysis uses a DualEncoder inspired by Jha et al. [2025]. Each survey is mapped through a model-specific MLP into a shared latent $Z{=}256$ and decoded back. Training combines reconstruction, RBF-MMD, cycle consistency, and Gram-matrix geometry preservation with equal weights (80 epochs, hidden width 512, two seeds). The two training sets are disjoint (5500 objects per survey), so the model never sees paired cross-survey correspondences.

Evaluation uses a held-out paired set ($n{=}2884$) and measures neighbourhood agreement between the translated representation and the true opposite-survey embedding. This is a different relational functional from the matched SAE/BSF comparison, so we use it to test the *shape* of the size dependence rather than to rank absolute mKNN values across methods.

3 Results

3.1 Structured probes reveal substantially more correspondence

Against the matched held-out dense baseline at $k{=}10$, SAE lifts every rung (mean $+0.0056$, range $+0.0025$–$+0.0127$), while BSF gives a larger mean lift of $+0.0076$ (range $+0.0017$–$+0.0168$). For example, ConvNeXt nano moves from 0.0092 dense to 0.0149 with SAE and 0.0161 with BSF; I-JEPA giant moves from 0.0176 to 0.0274 and 0.0321, respectively. The dense paper-style scores are already above chance (≈ 0.007–0.018, mean 0.0115), but placing the SAE and BSF codes in a shared basis exposes substantially more correspondence.

Figure 1 shows the important visual pattern: the probes produce large vertical shifts, while the within-family trajectories change much less.

3.2 The extra correspondence does not strengthen systematically with scale

Under the original paper-style dense evaluation, $8/11$ adjacent size steps are positive (one-sided $p{=}0.113$), which recovers the mild direction of the earlier astronomy scaling result on this Legacy-only subset. The main analysis, however, is the matched held-out comparison.

The 11 within-family probe×scale interactions cluster around zero (Figure 2b). For SAE, mean $D{=}{-}5.2{\times}10^{-4}$, median $+4.9{\times}10^{-4}$, range $[-4.6, +1.2]{\times}10^{-3}$, with 6 positive and 5 negative.

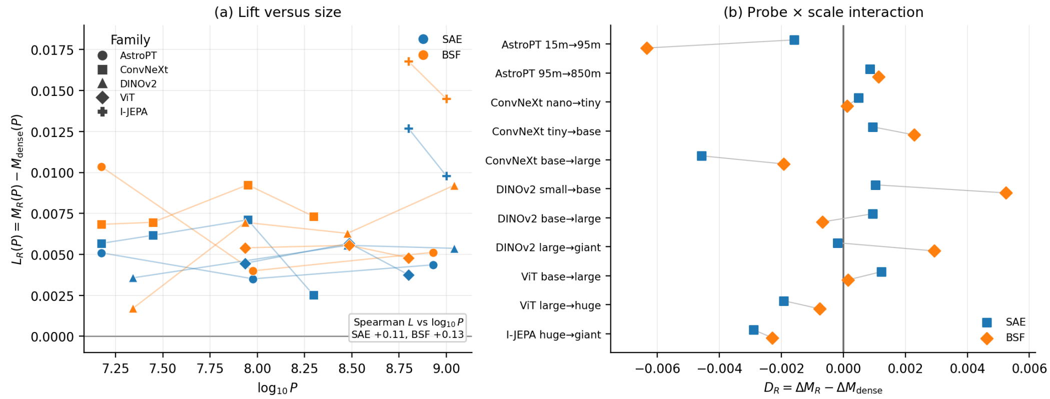

Figure 2: (a) Matched held-out lift $L_R(P)$ versus $\log_{10} P$. (b) Adjacent probe×scale interaction $D_R = \Delta M_R - \Delta M_{\mathrm{dense}}$ for the same 11 within-family steps. The lift is positive across the ladder, including for small models; its change with model size is small and mixed in sign.

For BSF, mean $D{\approx}0$, median $+1.2{\times}10^{-4}$, range $[-6.3, +5.2]{\times}10^{-3}$, again with 6 positive and 5 negative.

The same picture appears in the lift itself. Spearman correlation between $L_R(P)$ and $\log_{10} P$ is $+0.11$ for SAE and $+0.13$ for BSF. In other words, the shared basis exposes more transferable sparse structure, but that extra structure does not preferentially appear in the larger models. Increasing parameter count does not necessarily increase shared representation.

Raw adjacent-sign counts are consistent with this reading rather than being the main result. On the matched holdout they are $9/11$ for dense, $7/11$ for SAE, and $9/11$ for BSF. This is why it would be misleading to say that BSF "does not scale": it can rise with size while its *increment over dense* remains approximately size-independent.

3.3 Unpaired alignment gives the same qualitative size picture

The unpaired probe also recovers nontrivial relational structure. Its mKNN@10 scores lie around 0.012–0.023, compared with a random-neighbour floor of $10/2883 \approx 0.0035$, despite the model never seeing paired cross-survey objects during training. Its size trajectories are nevertheless non-monotonic.

For ConvNeXt, mKNN@10 follows

$$0.016 \to 0.012 \to 0.023 \to 0.012,$$

while DINOv2 gives

$$0.022 \to 0.019 \to 0.020 \to 0.019.$$

Only $2/6$ adjacent steps are positive. With six transitions, the trajectory itself is more informative than a significance test.

This does not put unpaired alignment on the same absolute scale as SAE or BSF. Its value is that the absence of clean monotonicity persists under a substantially different alignment construction: a larger model need not be easier to translate into a shared geometry.

4 Discussion and limitations

The main result is that model scaling does not necessarily imply shared representation. A learned shared basis makes substantial Legacy↔HSC correspondence visible at every rung, including the smallest, yet the extra correspondence does not grow systematically with parameter count. The unpaired result points in the same direction from a different construction. This is hard to dismiss as an artefact of dense coordinates: when we look for shared structure more carefully, we find more of it, but we do not find that bigger models own a larger share.

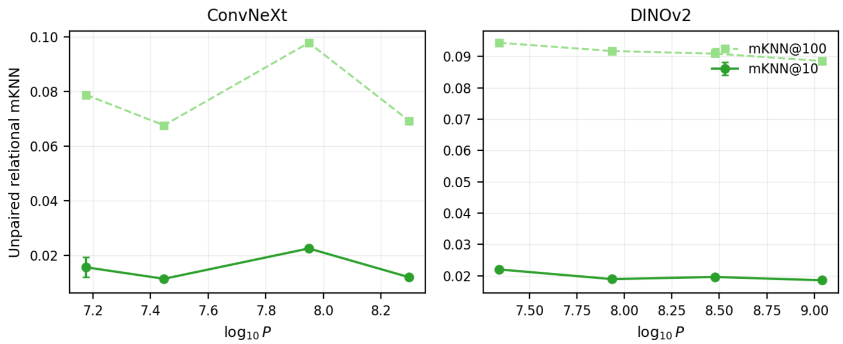

Figure 3: Unpaired DualEncoder relational mKNN versus $\log_{10} P$ (mean $\pm$ seed sd; two seeds). Solid: mKNN@10. Dashed: mKNN@100. The curves are non-monotonic across both available families.

One possibility is that raw parameter count is simply not the right variable. It is not an invariant measure of functional complexity: parameters can be added redundantly without materially changing the represented function. Model size may therefore be standing in for something more relevant, such as the pressure to achieve capability under constrained computational resources. Under that view, representational convergence could depend more directly on efficiency or compression pressure than on parameter count itself. We treat this as a hypothesis for future work. The connection to computationally relative notions of structure such as epiplexity [Finzi et al., 2026] is suggestive, but it is not needed for the present result.

A same-survey, cross-architecture analogue on Physics galaxies is reported in Appendix A. The shared basis again raises the level of correspondence on every pair, but those pairs are not a size ladder and are not used to estimate D_R.

The limitations are straightforward. There are only 11 paired adjacent transitions, and the unpaired analysis covers two families. SAE TopK sparsity is not perfectly homogeneous across the ladder ($k \in \{18, \ldots, 23\}$, with fixed F and seed). The unpaired analysis uses a different training and evaluation construction by design. Finally, learned alignment procedures can induce correspondence in ways that an analytical mKNN chance baseline does not capture; pipeline-refit permutation nulls are therefore a natural next control [Gröger et al., 2026].

5 Conclusion

A learned shared basis between independently trained SAE representations substantially increases measured cross-modal transfer relative to dense embeddings, including for small models. The additional correspondence it exposes does not systematically increase with model size. The same pattern holds for shared-basis BSF and for a qualitatively different unpaired relational probe. Putting sparse representations into a common basis strongly affects how much shared structure is visible; making the models larger does not reliably make that structure more shared.

The implication is straightforward: scaling a model does not necessarily make its representation more Platonic. The next question is what quantity actually governs representational convergence.

Acknowledgments

Code and exact configurations will be released with the camera-ready version.

References

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15619–15629, 2023. doi: 10.1109/CVPR52729.2023.01499.

Arjun Dey, David J. Schlegel, Dustin Lang, Robert Blum, Kaylan Burleigh, Xiaohui Fan, Joseph R. Findlay, Doug Finkbeiner, David Herrera, Stéphanie Juneau, et al. Overview of the DESI legacy imaging surveys. *The Astronomical Journal*, 157(5):168, 2019. doi: 10.3847/1538-3881/ab089d.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

Thomas Fel, Matthew Kowal, Mozes Jacobs, Dron Hazra, Usha Bhalla, Lee Sharkey, Lucius Bushnaq, Satchel Grant, Tal Haklay, Thomas Icard, Can Rager, Michael Pearce, Daniel Wurgaft, Aiden Swann, Fenil Doshi, Siddharth Boppana, Curt Tigges, Nick Cammarata, Thomas Serre, Vasudev Shyam, Owen Lewis, Thomas McGrath, Jack Merullo, Ekdeep Singh Lubana, and Atticus Geiger. Structuring sparsity: Block-sparse featurizers capture visual concept manifolds. *arXiv preprint arXiv:2606.25234*, 2026.

Marc Finzi, Shikai Qiu, Yiding Jiang, Pavel Izmailov, J. Zico Kolter, and Andrew Gordon Wilson. From entropy to epiplexity: Rethinking information for computationally bounded intelligence. *arXiv preprint arXiv:2601.03220*, 2026.

Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations*, 2025. URL `https://arxiv.org/abs/2406.04093`.

Fabian Gröger, Shuo Wen, and Maria Brbić. Revisiting the Platonic representation hypothesis: An Aristotelian view. *arXiv preprint arXiv:2602.14486*, 2026.

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The Platonic representation hypothesis. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 20617–20642. PMLR, 2024. URL `https://proceedings.mlr.press/v235/huh24a.html`.

Rishi Jha, Collin Zhang, Vitaly Shmatikov, and John X. Morris. Harnessing the universal geometry of embeddings. In *Advances in Neural Information Processing Systems*, 2025. URL `https://proceedings.neurips.cc/paper_files/paper/2025/hash/4175dee33d6145cb8f0323703d138a53-Abstract-Conference.html`.

Michael Lan, Philip Torr, Austin Meek, Ashkan Khakzar, David Krueger, and Fazl Barez. Quantifying feature space universality across large language models via sparse autoencoders. *arXiv preprint arXiv:2410.06981*, 2024.

Satoshi Miyazaki, Yutaka Komiyama, Satoshi Kawanomoto, Yukiko Fujimori, Takashi Hamana, Hisanori Hayashi, Hiroki Heiguchi, Takashi Hattori, Shiang-Yu Huang, Masatoshi Imanishi, et al. Hyper Suprime-Cam: System design and verification of image quality. *Publications of the Astronomical Society of Japan*, 70(SP1):S1, 2018. doi: 10.1093/pasj/psx063.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR, 2021. URL `https://proceedings.mlr.press/v139/radford21a.html`.

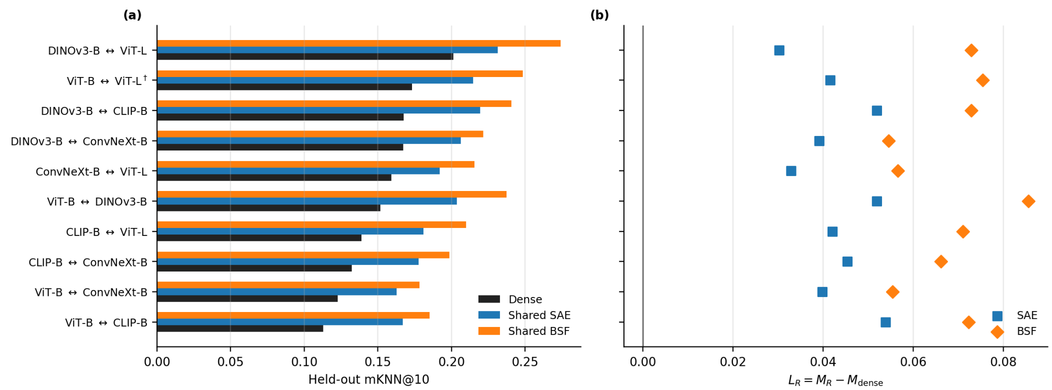

Figure 4: Physics galaxies, same objects, different models. (a) Held-out mKNN@10 for dense embeddings, SAE codes mapped into a shared dictionary, and the same protocol for BSF. (b) Lift $L_R = M_R - M_{\rm dense}$; all ten pairs are positive. †The only within-family size pair (ViT-B↔ViT-L). This figure supports the *level* claim only.

Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025.

Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling large observation models for astronomy. In *ICML 2024 AI for Science Workshop*, 2024.

UniverseTBD, Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The Platonic universe: Do foundation models see the same sky? *arXiv preprint arXiv:2509.19453*, 2025. doi: 10.48550/arXiv.2509.19453. Accepted to the NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences.

Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16133–16142, 2023.

A Cross-model shared-basis transfer

The main ladders pair the same architecture across two surveys. Here we ask the complementary *level* question: if two independently trained models see the *same* objects, does a learned shared basis still increase neighbourhood transfer between their SAE (and BSF) codes? This is not a probe×scale test. Nine of the ten pairs are cross-architecture comparisons at comparable capacity; only ViT-B↔ViT-L is a within-family size step, which is not enough to estimate D_R or a lift-versus-size correlation.

We use same-object Physics galaxy embeddings (n=16,384 subsample; $n_{\rm test}$=3277; k=10) spanning ViT [Dosovitskiy et al., 2021], ConvNeXt [Woo et al., 2023], CLIP [Radford et al., 2021], and DINOv3 [Siméoni et al., 2025]. The Ridge map is fit on the training split only. Unlike the main paper, the reported scores use `shared_best_cosine` (the better of the two mapping directions on the evaluation split), so SAE and BSF are comparable to each other and to dense *within this suite*, but they are not interchangeable with the predefined-side1 Legacy↔HSC numbers.

Dense cross-model alignment is already high (mean mKNN@10 0.153, range 0.113–0.201), an order of magnitude above the official Legacy↔HSC dense scores. The shared basis still adds a large, uniformly positive increment: SAE mean lift $+0.043$ (range $+0.030$–$+0.054$); BSF mean lift $+0.068$ (range $+0.055$–$+0.086$); $10/10$ pairs for both (Figure 4; Table 1). The single within-family pair (ViT-B↔ViT-L) is in line with the other nine (dense 0.173; SAE 0.215; BSF 0.249), not a scale trend.

Two unofficial same-survey ViT-B↔DINOv3 pairs on Legacy images sit in a different regime (HSC dense 0.0043; Legacy Survey dense 0.0038; lifts $\lesssim 0.0013$) and are not pooled with Physics.

Table 1: Physics holdout-20 cross-model mKNN@10 (n_{test}=3277). Mapping: `shared_best_cosine`. †Within-family size pair.

Pair	Dense	SAE	BSF
ViT-B ↔ CLIP-B	0.113	0.167	0.185
ViT-B ↔ ConvNeXt-B	0.123	0.163	0.178
CLIP-B ↔ ConvNeXt-B	0.132	0.178	0.199
CLIP-B ↔ ViT-L	0.139	0.181	0.210
ViT-B ↔ DINOv3-B	0.152	0.204	0.237
ConvNeXt-B ↔ ViT-L	0.159	0.192	0.216
DINOv3-B ↔ ConvNeXt-B	0.167	0.206	0.222
DINOv3-B ↔ CLIP-B	0.168	0.220	0.240
ViT-B ↔ ViT-L†	0.173	0.215	0.249
DINOv3-B ↔ ViT-L	0.201	0.231	0.274
Mean	0.153	0.196	0.221